

| Assessment | Test | Domain  | Skill                            | Difficulty                               |
|------------|------|---------|----------------------------------|------------------------------------------|
| SAT        | Math | Algebra | Linear equations in one variable | Hard <div></div> <div></div> <div></div> |

Question

Each side of a **30**-sided polygon has one of three lengths. The number of sides with length **8 centimeters (cm)** is **5** times the number of sides  $n$  with length **3 cm**. There are **6** sides with length **4 cm**. Which equation must be true for the value of  $n$ ?

Answer

- A.  $5n + 6 = 30$
- B.  $6n + 6 = 30$
- C.  $8n + 3n + 4n = 30$
- D.  $8(5n) + 3n + 4(6) = 30$

Correct Answer: B

Rationale

Choice B is correct. It's given that each side of a **30**-sided polygon has one of three lengths. It's also given that the number of sides with length **8 centimeters (cm)** is **5** times the number of sides  $n$  with length **3 cm**. Therefore, there are  $5 \times n$ , or  $5n$ , sides with length **8 cm**. It's also given that there are **6** sides with length **4 cm**. Therefore, the number of **3 cm**, **4 cm**, and **8 cm** sides are  $n$ , **6**, and  $5n$ , respectively. Since there are a total of **30** sides, the equation  $n + 6 + 5n = 30$  represents this situation. Combining like terms on the left-hand side of this equation yields  $6n + 6 = 30$ . Therefore, the equation that must be true for the value of  $n$  is  $6n + 6 = 30$ .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.